Reverse Customer Name

Week 1 Practice - Problem 5 | Java

Problem Summary

Return a reversed copy of the customer name and print both original and reversed names without modifying the original string.

Java Solution

```
public class Main {
    static String reverseCustomerName(String customerName) {
        char[] chars = customerName.toCharArray();
        int left = 0, right = chars.length - 1;
        while (left < right) {
            char temp = chars[left];
            chars[left] = chars[right];
            chars[right] = temp;
            left++; right--;
        }
        return new String(chars);
    }
    public static void main(String[] args) {
        String customerName = "Sunil";
        System.out.println("Original Name: " + customerName);
        System.out.println("Reversed Name: " + reverseCustomerName(customerName));
    }
}
```

Sample Output

```
Original Name: Sunil
Reversed Name: linuS
```

Complexity / Key Notes

Time: O(n). Extra space: O(n) for the array and the returned string. Java String is immutable, so the original name remains unchanged.